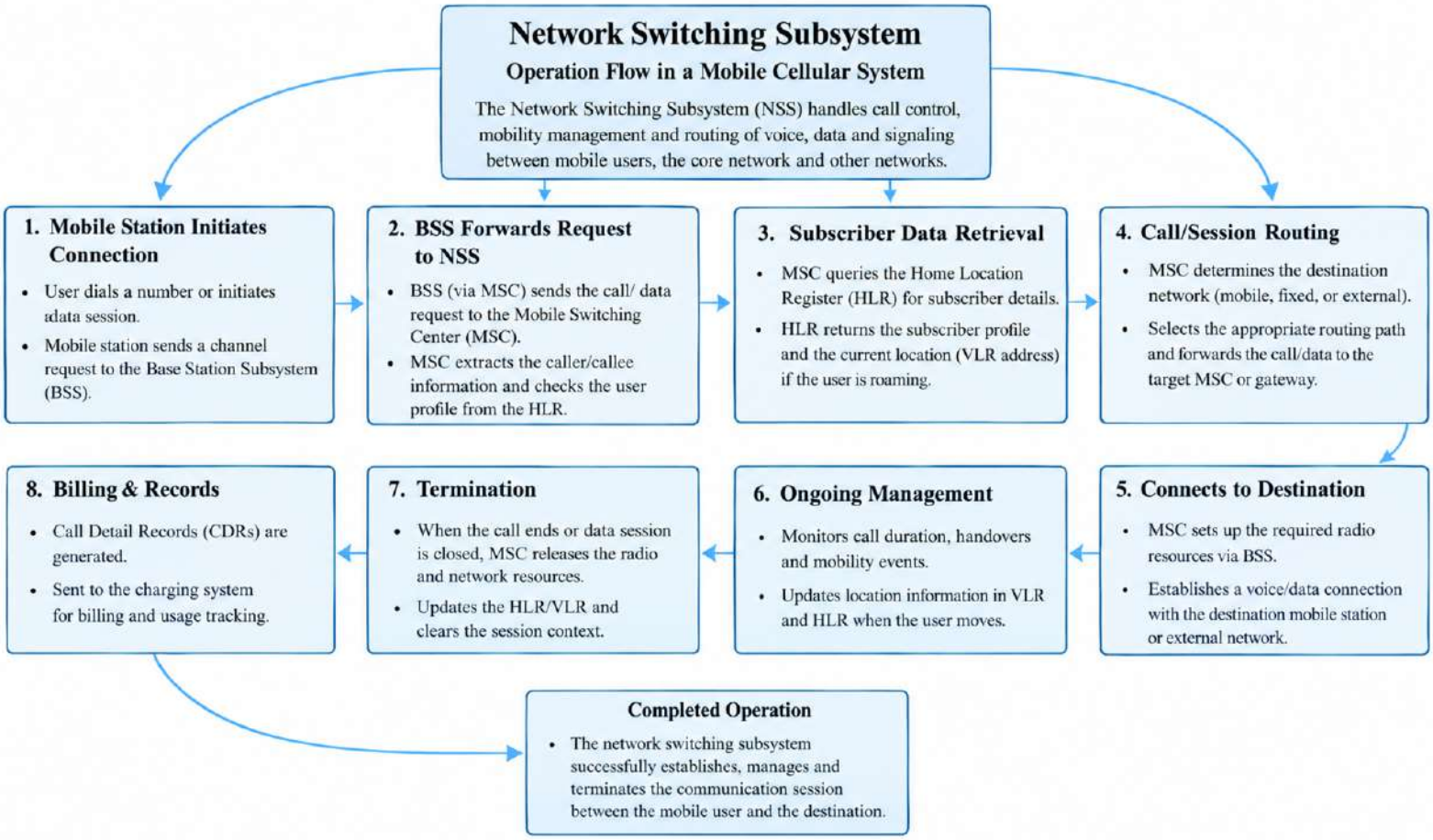

Network Switching Subsystem
Operation Flow in a Mobile Cellular System
The Network Switching Subsystem (NSS) handles call control, mobility management and routing of voice, data and signaling between mobile users, the core network and other networks.
1. Mobile Station Initiates Connection
User dials a number or initiates adata session.
Mobile station sends a channel request to the Base Station Subsystem (BSS).
2. BSS Forwards Request to NSS
BSS (via MSC) sends the call/ data request to the Mobile Switching Center (MSC).
MSC extracts the caller/callee information and checks the user profile from the HLR.
3. Subscriber Data Retrieval
MSC queries the Home Location Register (HLR) for subscriber details.
HLR returns the subscriber profile and the current location (VLR address) if the user is roaming.
4. Call/Session Routing
MSC determines the destination network (mobile, fixed, or external).
Selects the appropriate routing path and forwards the call/data to the target MSC or gateway.
5. Connects to Destination
MSC sets up the required radio resources via BSS.
Establishes a voice/data connection with the destination mobile station or external network.
6. Ongoing Management
Monitors call duration, handovers and mobility events.
Updates location information in VLR and HLR when the user moves.
7. Termination
When the call ends or data session is closed, MSC releases the radio and network resources.
Updates the HLR/VLR and clears the session context.
8. Billing & Records
Call Detail Records (CDRs) are generated.
Sent to the charging system for billing and usage tracking.
Completed Operation
The network switching subsystem successfully establishes, manages and terminates the communication session between the mobile user and the destination.